LUKAS JOSÉ FERRER

lukasjferrer@gmail.com · 310.801.8751 · Los Angeles, CA

Product-minded ML engineer with 7+ years building production AI systems from zero to one at a healthcare startup serving Fortune 100 customers. Architected a fault-tolerant medical-grade NLP ETL pipeline that has chewed through 100M+ pages of clinical records in production at $0.0269/page — plus the multi-stage LLM and custom Transformer systems sitting on top of it that turn raw documents into physician-ready case summaries with citation validation and PHI protection. Comfortable annealing noisy user-generated training data into custom models that hit real business KPIs, then welding them onto frontier APIs (Anthropic, OpenAI). Has been productizing LLM APIs alongside brewed-in-house predictive models since the day ChatGPT shipped — knows the ins-and-outs of tool use, what to actually expect from a given model, and how to make messy unstructured data purr. Operates on the frontier of agentic coding — turns AI coding agents (Claude Code, sub-agents, MCP-driven workflows) into shippable products that can make a fully bootstrapped company profitable, not demos, by managing to verified outcomes and token cost rather than engineer-hours and shipping at multiple-engineer velocity.

EXPERIENCE

Lead ML & DevOps Engineer — Sequoia Port Inc.

Mar 2021 – Present · Remote — Los Angeles, CA

Healthcare Products & Clinical AI

- Reduced insurance claim review time by 65% and decreased specialist escalations by 40%, enabling healthcare clients to save millions on internal medical record reviews.
- Built an automated sub-record segmentation product that organized 1,000+ page medical cases for physician review in under an hour — replacing days of manual work and directly improving clinician workflows.
- Mined the NLP ETL pipeline's structured outputs to build a domain ontology and Graph DB on top of it — lifting RAG retrieval from 87% to 95% Recall@5, a level Semantic Search and Query Expansion alone could not reach, and powering chat-based case discovery and visualization features for clinical users on the frontend.
- Maintained HIPAA-compliant data pipelines with PHI anonymization (fake record ID systems), encryption at rest and in transit, and audit logging across all processing stages.
- Implemented PATE (Private Aggregation of Teacher Ensembles) to aggregate sensitive, user-generated training data across healthcare customers with differential privacy guarantees — enabling a single model to serve all clients with rising accuracy and without exposing any one customer's proprietary medical data.

LLM Pipeline Engineering

- Designed CaseOverviewV3: a multi-stage LLM pipeline (7 Lambda steps, 8 parallel summary categories) that generates comprehensive executive summaries from medical cases — producing physician-ready clinical documentation with structured narrative output.
- Built a citation validation system with fake-ID anonymization for PHI protection, hallucinated reference detection, and automatic retry with temperature escalation — ensuring LLM outputs only reference real source documents.
- Optimized LLM pipeline cost and latency through Anthropic prompt caching (~90% cost reduction on repetitive extractions), DynamoDB response caching (14-day TTL), per-task model tier selection, and aggressive token budgeting to minimize input costs.
- Implemented multi-level parallelization: distributed MAP state machines processing 8 summary categories in parallel, Lambda-level multiprocessing for concurrent LLM calls, and batch processing with configurable limits.
- Shipped both custom-trained models (LoRAs, distilled Transformers, ELECTRA-pretrained classifiers) and frontier-API features (Anthropic, OpenAI) into production — wrangling noisy, user-generated training data and iteratively annealing target labels and offline metrics toward the real business KPIs that pay the bills (specialist escalation rate, reviewer hours saved, claim review time).

Production ML & Infrastructure

- Architected the production medical-grade NLP ETL backend serving Fortune 100 healthcare customers — 100M+ pages processed (98% of all company throughput), fault-tolerant orchestration via AWS Step Functions with retries, dead-letter queues, and idempotent stages, and end-to-end unit economics driven down to $0.0269/page.
- Designed a 2-stage multimodal Transformer architecture (page images + OCR text + layout features) for sub-document segmentation, achieving 85% accuracy and >0.95 precision on medical relevancy filtering.
- Engineered a 2-step serving pipeline (embed all pages → classify all windows) achieving 5M+ pages/month throughput on 2–4 GPUs, optimizing batch processing and memory management for cost-effective inference.

- Implemented full CI/CD via AWS CDK (infrastructure-as-code) and wrote all shared utilities as reusable Lambda layers (AWS clients, OCR, LLM caller, serverless DB access) — used across 10+ production services.

Red Team (Contract) — OpenAI

Jan 2024 – Jun 2024 · Remote — Los Angeles, CA

- Contracted red-teamer on GPT-4o, probing the model's failure to express calibrated uncertainty in clinical contexts. Designed an eval where the model assigned ICD-10 codes and rationales to synthetically generated medical conditions, then jointly analyzed the extracted condition, the assigned code, and the rationale. Surfaced a class of confidently-fabricated outputs — codes that satisfied the regex for real ICD-10s but didn't actually exist in the ontology — and demonstrated that the failure mode is only remediable by grounding the model in the ICD-10 ontology via a RAG system (the limits of which were the test).

Regulatory Software Engineer — Neuralink

Oct 2020 – Jan 2021 · Fremont, CA

- Joined to support FDA regulatory preparation for a Class III implantable brain-computer interface — identified noncompliance in medical device design workflows and proposed remediation strategies aligned with FDA requirements while minimizing disruption to engineering productivity. Left the role following leadership's decision to pursue a go-forward approach that deprioritized Class III requirements in prototyping, rendering both much of the role's original scope, and the remediation strategy provided, moot. Can provide further detail as appropriate and within the bounds of the non-disclosure agreement.

Research Programmer — Information Sciences Institute (USC)

May 2018 – Oct 2020 · Marina del Rey, CA

- Worked as the top contributor on ISI's Neural Machine Translation training pipeline (under its principal architect) — integrated Transformer architectures straight out of academic papers, trained thousands of translation models before the "L" in LLM stood for anything, and pushed BLEU scores to compete with Google Translate across several rare languages. The reps are why I know what's actually going on under the hood instead of just at the API surface.
- Developed an automated ETL pipeline ingesting 100,000+ news articles/day from LexisNexis for ISI's SAGE geopolitical forecasting platform — feeding both a semantic search system and a generative Transformer sitting on top of it, and lifting relevant document retrieval by 73% (Recall@5). Effectively RAG before the community had a name for it.

EDUCATION

M.S. Biomedical Engineering — USC Viterbi School of Engineering, August 2017

Focus: Electrophysiology, Medical Device Regulations, Agile Software Development

B.S. Biomedical Engineering — USC Viterbi School of Engineering, May 2016

Focus: Medical Devices & Physiological Modeling

TECHNICAL SKILLS

Healthcare AI & Compliance: Clinical Document Processing, Medical Record ETL, PHI Anonymization, HIPAA-Compliant Pipelines, PATE (Differential Privacy), FDA Regulatory Workflows, Health Record Segmentation & Classification

LLM Systems: Anthropic API, OpenAI API, Multi-Stage LLM Pipelines, Agentic Coding (Claude Code, Cursor, MCP, Sub-Agents, Tool Use), Prompt Engineering, Citation Validation, Hallucination Detection, Prompt Caching, Model Orchestration, LoRA Fine-Tuning, Model Distillation

ML & Deep Learning: PyTorch, Custom Transformer Architectures, Multimodal Models, ELECTRA Pretraining, Contrastive Learning, Hugging Face Transformers, FAISS, NER, Document Classification

Infrastructure: AWS (CDK, Lambda, Step Functions, EC2, ECS, RDS, S3, SQS, CloudWatch, DynamoDB), Docker, CI/CD, Linux, SQL, VectorDBs, Graph Databases

Languages & Web: Python, TypeScript, React, Ruby on Rails, REST APIs, SQL, Bash